REVIEW SUMMARY

SEISMOLOGY

Deep-learning seismology

S. Mostafa Mousavi* and Gregory C. Beroza

BACKGROUND: Seismology is the study of seismic waves to understand their origin—most obviously, sudden fault slip in earthquakes, but also explosions, volcanic eruptions, glaciers, landslides, ocean waves, vehicular traffic, aircraft, trains, wind, air guns, and thunderstorms, for example. Seismology uses those same waves to infer the structure and properties of planetary interiors. Because sources can generate waves at any time, seismic ground motion is recorded continuously, at typical sampling rates of 100 points per second, for three components of motion, and on arrays that can include thousands of sensors. Although seismology is clearly a data-rich science, it often is a data-driven science as well, with new phenomena and unexpected behavior discovered with regularity. And for at least some tasks, the careful and painstaking work of seismic analysts over decades and around the world has also made seismology a data label–rich science. This facet makes it fertile ground for deep learning, which has entered almost every subfield of seismology and outperforms classical approaches, often dramatically, for many seismological tasks.

ADVANCES: Seismic wave identification and onset-time, first-break determination for seismic *P* and *S* waves within continuous seismic data are foundational to seismology and are particularly well suited to deep learning because of the availability of massive, labeled datasets. It has received particularly close attention, and that has led, for example, to the development of deep learning–based earthquake catalogs that can feature more than an order of magnitude more events than are present in conventional catalogs. Deep learning has shown the ability to outperform classical approaches for other important seismological tasks as well, including the discrimination of earthquakes from explosions and other sources, separation of seismic signals from background noise, seismic image processing and interpretation, and Earth model inversion.

OUTLOOK: The development of increasingly cost-effective sensors and emerging ground-motion sensing technologies, such as fiber optic cable and accelerometers in smart devices, portend a continuing acceleration of seismological data volumes, so that deep learning is likely to become essential to seismology's future. Deep learning's nonlinear mapping ability, sequential data modeling, automatic feature extraction, dimensionality reduction, and reparameterization are all advantageous for processing high-dimensional seismic data, particularly because those data are noisy and, from the point of view of mathematical inference, incomplete. Deep learning for scientific discovery and direct extraction of insight into seismological processes is clearly just getting started.

Aspects of seismology pose interesting additional challenges for deep learning. Many of the most important problems in earthquake seismology—such as earthquake forecasting, ground motion prediction, and rapid earthquake alerting—concern large and damaging earthquakes that are (fortunately) rare. That rarity poses a fundamental challenge for the data-hungry methods of deep learning: How can we train reliable models, and how do we validate them well enough to rely on them when data are scarce and opportunities to test models are infrequent? Further, how can we operationalize deep-learning techniques in such a situation, when the mechanisms by which they make predictions from data may not be easily explained, and the consequences of incorrect models are high? Incorporating domain knowledge through physics-based and explainable deep learning and setting up standard benchmarking and evaluation protocols will help ensure progress, as is the nascent emergence of a seismological data science ecosystem. More generally, a combination of data science literacy for geoscientists as well as recruiting data science expertise will help to ensure that deep-learning seismology reaches its full potential. ■

The list of author affiliations is available in the full article online.
*Corresponding author. Email: mousavim@google.com
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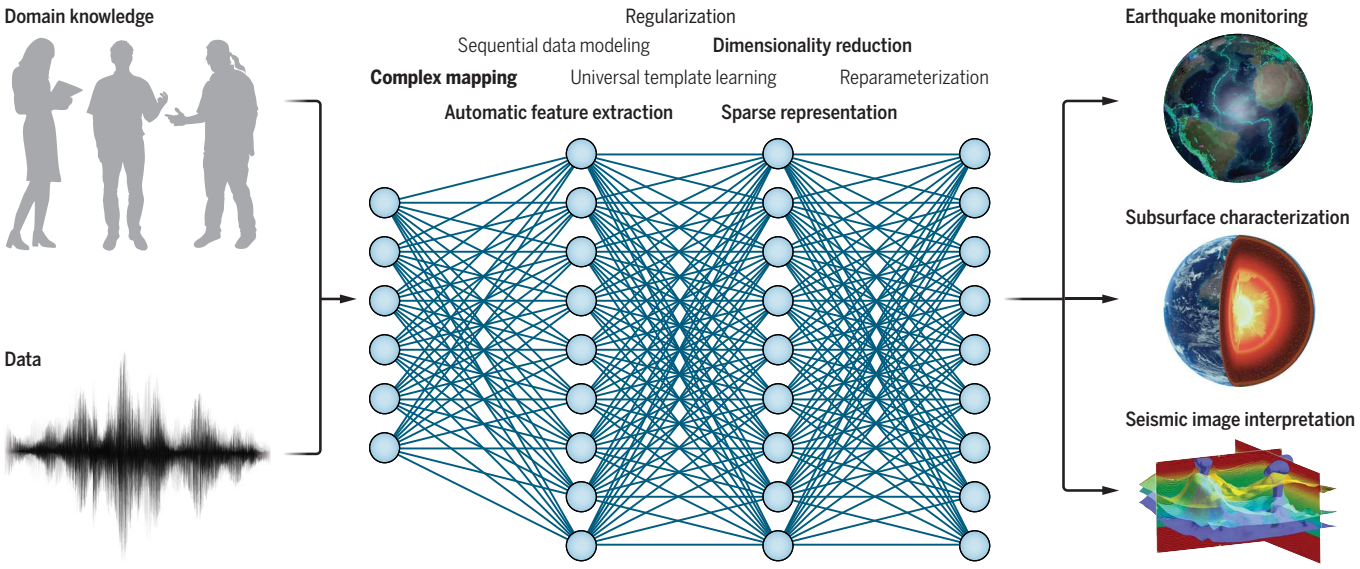

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Deep-learning processing of seismic data and incorporation of domain knowledge can lead to new capabilities and new insights across seismology.

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REVIEW

SEISMOLOGY

Deep-learning seismology

S. Mostafa Mousavi[1,2]* and Gregory C. Beroza[1]

Seismic waves from earthquakes and other sources are used to infer the structure and properties of Earth's interior. The availability of large-scale seismic datasets and the suitability of deep-learning techniques for seismic data processing have pushed deep learning to the forefront of fundamental, long-standing research investigations in seismology. However, some aspects of applying deep learning to seismology are likely to prove instructive for the geosciences, and perhaps other research areas more broadly. Deep learning is a powerful approach, but there are subtleties and nuances in its application. We present a systematic overview of trends, challenges, and opportunities in applications of deep-learning methods in seismology.

Seismology is a data-rich and data-driven science in which the application of deep-learning techniques is growing rapidly (Fig. 1). Deep learning (*1*) uses deep neural networks (DNNs) to learn complex relationships, find multiscale structure and patterns, and address ill-posed inverse problems by developing high-level representations of data using stacked layers of neurons and multiple nonlinear transformations. The abilities of DNN variants—such as convolutional neural networks (CNNs) to extract multiscale features from data and recurrent neural networks (RNNs) to model temporal relations in sequential data—make them powerful tools for seismology.

Deep-learning approaches have entered almost every subfield of seismology, for which they have shown the ability to outperform classical approaches, often dramatically, for seismological tasks such as denoising, earthquake detection, phase picking, seismic image processing and interpretation, and inverse and forward modeling. Some properties of DNNs—such as their universal approximation capability, automatic feature extraction, and dimensionality reduction—have been shown to be particularly advantageous in processing high-dimensional seismic recordings, which often are noisy and incomplete (Fig. 2). Seismological deep learning can process massive amounts of multifidelity seismic observations with unprecedented spatiotemporal coverage and lead to new insights and discoveries (*2*). Deep learning may be particularly effective for seismological problems for which the underlying physical processes are incompletely understood but for which the data are abundant and of high quality.

We present a systematic overview of deep-learning methods in seismology by sampling the current state of the art and through a meta-analysis of 637 journal papers published between January 1988 and January 2022 (*3*). We conclude with some speculation on future trends and with recommendations intended to accelerate progress.

[1]Department of Geophysics, Stanford University, Stanford, CA 94305, USA. [2]Google, Mountain View, CA 94043, USA.
*Corresponding author. Email: mousavim@google.com

Data processing automation

Seismic data are recorded (often irregularly or heterogeneously) as time series of ground motion by sensors that are deployed either sparsely (in a network) or densely (in an array) and that register acceleration, velocity, or displacement as output. Typically, these quantities are recorded in three perpendicular directions, so that we work with three-component vector ground motion. Important extensions to conventional seismic recording include rotational motion (*4*) and tensorial strain (*5*).

In both passive and active-source seismology, usually a sequence of processing and analyzing steps exists that needs to be performed. Despite decades of earlier efforts to develop algorithms to automate such data-processing and -analyzing tasks, many cases remain for which manual processing by skilled analysts is the most reliable option (such as phase picking or seismic interpretation). The accelerating expansion of seismic data volumes poses new challenges and brings with it the need to develop a new generation of robust processing tools by using data-driven approaches such as deep learning.

Seismic event monitoring

The classification capability of DNNs makes them ideal for processing tasks such as event discrimination (*6*–*8*), event detection (*9*, *10*), phase picking (*11*, *12*), first-polarity determination (*13*, *14*), and phase association (*15*, *16*) that together form a seismicity monitoring workflow.

Seismic event discrimination

Various studies have shown outstanding performance of deep learning for different event discrimination tasks, such as differentiating between human-made explosions and natural earthquakes (*6*, *7*), volcano-tectonic earthquakes and volcanic tremor (*8*), mining-related events (*17*), tectonic tremor and local earthquakes (*18*), and teleseismic and local earthquakes (*19*). The main challenge in developing discriminator models is the scarcity of large-scale labeled datasets. Efforts to address this include semisupervised learning (*20*), unsupervised learning (*19*), and transfer learning (*8*).

Earthquake detection and phase picking

Identification of earthquake signals among nonstationary background noise and a variety

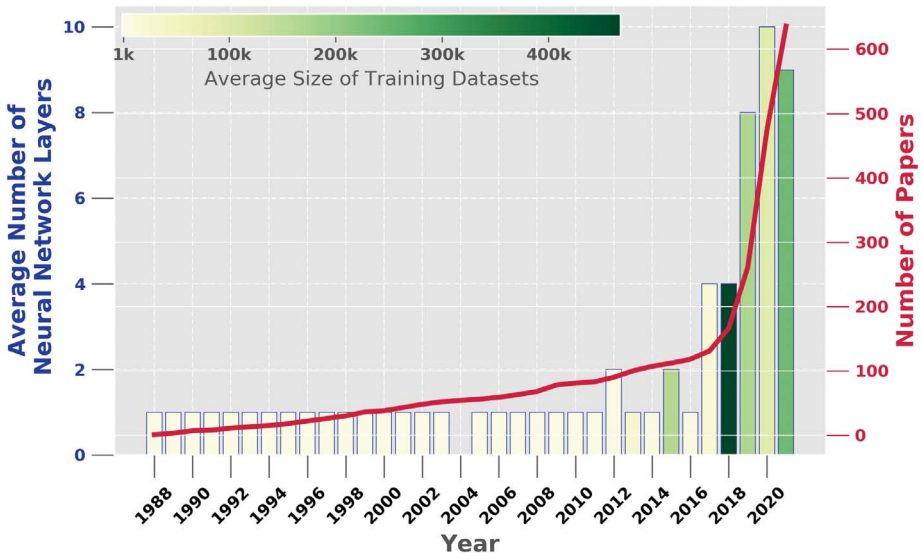

Fig. 1. A surge of interest in deep-learning applications in seismology. Cumulative number of published papers in seismology between January 1988 and January 2022 that used neural networks (red curve). Bars show the average number of hidden layers of neural networks—the depth of networks. The color scale depicts the average size of training data used to train the networks.

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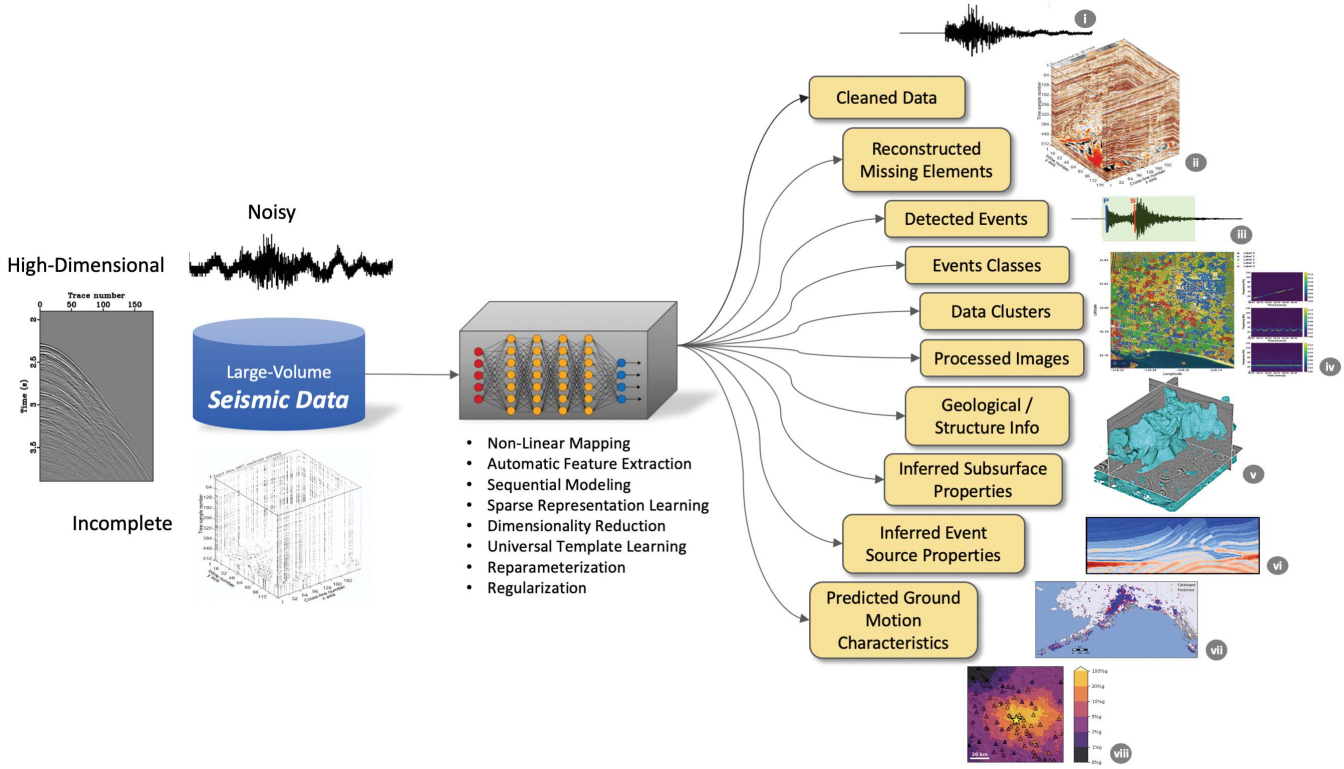

Fig. 2. The properties of DNNs for the processing of seismic data. Various properties of DNNs such as their universal approximation capability, automatic feature extraction, and dimensionality reduction are advantageous for processing large-volume, high-dimension seismic recordings, which often are noisy and incomplete. These attributes are used to build powerful models for various seismological tasks such as (i) seismic data denoising, (ii) interpolation of missing traces, (iii) earthquake signal detection and phase picking [reprinted with permission from (43)], (iv) exploratory data analysis [reprinted with permission from (123), copyright Seismological Society of America], (v) salt body detection, (vii) seismic velocity model building, (vii) earthquake location estimation, or (viii) ground motion prediction.

of nonearthquake signals, from both natural and anthropogenic sources, as well as precise measurement of *P* wave or *S* wave arrival times, is a challenging task—we differentiate between discrimination and the detection task in which the goal is to identify only one signal of interest among a variety of recorded events. The huge archives of seismic data and the availability of hand-picked labels from earthquake catalogs made these tasks to be among the earliest and the most popular applications of data-hungry deep learning in seismology. The potential of deep learning for these tasks was evident from the earliest deep learning–based studies (*9*–*11*, *21*). Since then, more than 80 models have been developed to explore the performance gain obtained through more complex deep-learning architectures, such as attentive CNN-RNN (*12*, *22*) and Capsule Networks (*23*). Researchers have also explored the advantages and disadvantages of performing these tasks in the time domain (*10*, *11*), the time-frequency domain (*21*), and a fusion of time and time-frequency (*24*); of using multistation data (*25*, *26*) and multisensor data (GPS and seismic); of using two observation levels (seismic station level and network level) (*27*); and of performing the two tasks sequentially (*28*, *29*) or simultaneously (*12*), on data recorded with low-cost microelectromechanical systems (MEMS) accelerometers (*30*), distributed acoustic sensing (DAS) (*31*), or ocean-bottom seismometers (OBSs), and geophones (*25*).

Deep-learning detectors and phase-pickers can be thought of as universal templates of earthquake signals represented by a set of kernels of different sizes at different layers of a neural network and the connection weights between them. The outstanding performance of these models comes from their ability to learn a deep and compact representation of diverse templates; however, because of the lack of standardized benchmarking, the optimal deep-learning approach for these tasks is unclear. Nevertheless, common observations and conclusions are that (i) deep-learning models systematically outperform classical characteristic function–based models; (ii) deep-learning model performance approaches that of template-matching methods, but with much lower computational cost and no need for template events; (iii) deep-learning models can pick phase-arrival times with a similar picking error to that of skilled analysts while picking far more phases in much less time; (iv) the power of artificial intelligence (AI)–based pickers is most evident for the challenging task of picking *S* waves that emerge from *P* coda; (v) deep-learning models can generalize well to other regions or data types beyond their training data; and (vi) compared with the classical methods, deep-learning models perform better in conditions with lower signal-to-noise ratio (SNR). As a result, deep-learning models provide more complete earthquake catalogs and much-higher-resolution maps of seismicity by detecting more small events.

Seismic image processing

Reflected and refracted seismic waves recorded by an array of seismic sensors are used to image subsurface geology and structure. Deep learning has proven an effective tool in the processing steps used to improve the quality of seismic images and to transform them into an interpretable image of the subsurface by removing data acquisition artifacts and wave propagation effects to highlight events that more accurately portray the true geology and structure.

Seismic denoising

Recorded seismic data are always contaminated by unwanted background noise or interference from other signals. This unwanted energy is often nonstationary with incoherent (random or Gaussian) and/or coherent components

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that can mask the signal of interest and affect subsequent inversion or interpretation. Seismic denoising (32) and closely related tasks such as multiple removal (33) and deblending (34) are a focus area for the applications of deep learning in seismology. These studies have shown the promising performance of deep learning for suppressing a wide variety of noise in seismic data, including the random noise (35), coherent low-frequency noise (36), swell noise (37), ground roll (38), and coherent and incoherent noise (32) in conventional and unconventional seismic data such as DAS (39) and in one-dimensional (1D) (32), 2D (40), and 3D (41) recordings. The effectiveness of deep-learning models in suppressing various types and levels of seismic noise with low computational cost is remarkable.

In deep-learning denoising, the statistics of signal and noise are directly learned from training data. As a result, the learned models achieve superior performance and are more flexible than conventional methods that rely on prior assumptions about the statistics of the noise or signal. The out-of-distribution generalization of their superior performance, however, depends highly on how well their training data represent the distributions of field data. A challenge in building training data for deep-learning denoisers is to obtain noise-free signals and signal-free noise from field recordings. An effective approach to meet this challenge is to create semisynthetic data in which clean synthetic signals are combined with manually picked noise from the field data (36, 39). Performing unsupervised denoising by using either autoencoders (35, 40) or generative adversarial networks (GANs) (36, 38) are other common approaches. The idea behind autoencoder denoisers is that the most representative features of the data that are automatically extracted by the autoencoder have a more coherent structure and represent signals of interest. Hence, autoencoders are usually only effective for random noise suppression. Generative models, on the other hand, learn the domain mapping from noisy data domain into the signal domain and could work for incoherent noise as well, but require examples of clean and noisy data.

Seismic image acquisition and processing

Tasks under this heading include identification of shots and receivers with position errors (42), interpolation for reconstructing traces missing during acquisition (43), extrapolation to low-frequency components on the basis of high-frequency energy (44) or vice versa (45), identification of the first arriving seismic energy in time-space domain (46), automation of velocity picking (47), extraction and classification of dispersion curves (48), and seismic migration. Although velocity picking and migration can be treated as an inversion process, we discuss them here under seismic image processing. Tasks in which the main goal is the inversion for some physical properties are discussed in a subsequent section (49, 50). The results of these studies suggest impressive capabilities of DNNs in performing subtle and complex processing tasks with unprecedented computational efficiency. For example, deep-learning seismic interpolation methods can successfully reconstruct missing regularly or irregularly spaced data with flexibility similar to that of traditional methods based on wave equation and signal processing, but without their high computational costs or limiting assumptions such as sparsity or low rank. The efficiency gain is evident from applications for seismic migration. Using a CNN, Liu *et al.* (49) accelerated prestack time migration by more than 100 times, and Kaur *et al.* (50) improved both computational efficiency and image resolution in least-squares migration by estimating the inverse Hessian operator using a cyclic GAN and eliminating the need for iteration.

Seismic image interpretation

Seismic image interpretation aims to extract useful information regarding subsurface geology (such as rock type) or structure (such as fault, horizon, and salt body) from seismic images that have limited resolution and are contaminated with noise. Many deep-learning techniques have been developed to automate and assist seismic-image interpretations that are subjective and often require substantial time and domain knowledge from the interpreter.

The feature-learning and dimensionality-reduction abilities of CNNs are well suited for clustering of seismic response in terms of geological properties or facies (51, 52) and outperform machine-learning approaches for seismic facies mapping based on hand-engineered features. Deep-learning models, however, have the potential to go beyond simple pattern recognition to provide accurate end-to-end classification results (53–55) while quantifying the uncertainties (56).

Besides geological interpretation, the power of deep learning for image segmentation and object detection makes it a popular choice for structural-interpretation tasks such as automatic identification of geological channels (57), bright spots (58), collapsed karst features (59), scatterers (60), bottom-simulating reflections (61), salt bodies (62), horizons (63, 64), and faults (65–67) in seismic images.

Seismic image interpretation is a type of image understanding. Hence, applications of popular neural network architectures developed for computer vision are a natural choice. The remarkable performance of these methods for seismic image interpretation—despite the scarcity of large, complete, and precisely labeled training datasets—is attributable to a combination of the extensive use of data augmentation and the generation of sophisticated synthetic data.

Forward problems

The impressive predictive power of DNNs makes them popular tools for forward modeling problems in seismology. We outline three categories—seismic wave simulation, ground motion characterization, and earthquake forecasting—as instructive examples of forward-modeling tasks in seismology for which DNNs have been used.

Seismic wave simulation

To date, applications of deep learning for seismic wave simulation have been limited. Nevertheless, deep learning appears to provide an effective alternative for standard numerical methods and can address issues such as discretization errors and high computational complexity. The use of physics-informed neural networks (PINNs) (68–71) and GANs (72–74) are two current trends. In PINNs, a DNN is trained to learn the solution of the wave equation for a medium by implicitly defining the boundary conditions and the wave equation in the loss function used in training of the neural network. These deep-learning models can learn to solve the wave equation in 2D or 3D media, even for complex faulted structure or topography, and generalize well beyond the time stamps of their training dataset. They are much more efficient in computing arbitrary space-time points in the wavefield than are traditional numerical simulations and reduce computation time by at least an order of magnitude.

GAN simulators use the universal function approximation ability of DNNs to learn the probability distributions of attributes of training data by optimizing a generator network. This generator model can be used as reparameterization of such distributions to generate new samples drawn from the learned distributions during inference. GANs have been used to generate synthetic earthquake and non-earthquake seismograms as a data augmentation tool for training deep-learning earthquake detectors (72); to generate broadband seismic signals by blending the low-frequency output of numerical physics-based simulations with sparsely sampled broadband observations (73); and to generate 3C strong motion time series for different magnitude, distance, and site conditions (74). They provide an efficient framework for generating large-scale synthetic training data to improve the performance of deep-learning classifiers and detectors.

Ground motion attribute modeling

Another category of forward models provides estimates of distinct attributes of ground motion caused by a seismic source such as peak ground acceleration (PGA) (75, 76) or

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the travel time of a seismic wave between two points in a medium (77).

Neural networks have been used to build nonparametric ground-motion models (GMMs) based on a fixed set of input parameters by using field (78) or synthetic (79) data. This group of AI-based GMMs (which is more than 15 years old) has similar input-output and performance to those of traditional methods; however, another group of deep learning–based GMMs introduces a more progressive generation of GMMs (75, 76, 80) under a new paradigm. These models can provide a rapid estimation of ground shaking directly from observed waveforms without an interpretation through earthquake source parameters or site conditions. A network-level pattern of ground-motion propagation contains enough information for a DNN to nowcast PGA at stations that have not been shaken by the earthquake waves yet, on the basis of those stations that have been shaken (76, 80). This group of end-to-end approaches seems particularly relevant for earthquake early warning (EEW).

Deep learning–based Eikonal solvers (77) have much lower space and time complexity compared with those of the traditional finite-difference approaches (especially for complex structures) because they eliminate the need for travel-time lookup tables and greatly benefit from the parallel computing capability of deep-learning platforms.

Earthquake forecasting

Solving the earthquake forecasting puzzle by using neural networks is an appealing topic (81). Early attempts relied on hand-engineered features, but the emphasis has shifted to neural network–based forecasting approaches that more directly incorporate the complete spatiotemporal structure of seismic catalogs (82, 83). This exploits the automatic feature extraction ability of some DNNs along with their potential for learning the dynamics of data to take a fresh look at an old problem. Most of these studies are in an experimental phase and focus on comparing the relative performance of (usually not fine tuned) different deep-learning and/or classical machine-learning predictors. It remains to be established whether deep-learning earthquake forecasting models offer new insights into earthquake predictability and whether they perform better than traditional empirical approaches (84).

The rapid progress in deep learning–based earthquake monitoring has resulted in deeper earthquake catalogs with unprecedented spatial resolution and magnitude of completeness that should improve both traditional statistical and physics-based forecasting methods (85) as well as AI-based approaches. Here too, the predictive value of newly uncovered small earthquakes in deep learning–based catalogs needs to be clarified (86). A key challenge is to effectively explore these complex and denser catalogs and extract new relationships. This is a task that deep learning can help with as well. For example, Zhu *et al.* (87) present a reinforcement-learning approach to model complex spatiotemporal relationships in earthquake catalogs that are not possible to model by using classical approaches such as the epidemic-type aftershock sequences (ETAS) method.

Deep learning offers new opportunities in earthquake forecasting by blending different potential precursory data and known earthquake physics with seismicity data and underlying physics that remain unknown. The growing number of forecasting competitions such as in (88), richterx (89), or AETA (90) is important for engaging broader contributions beyond the earthquake seismology community. More effort in setting up the technical rigor and benchmarking practices is needed to direct this new wave of enthusiasm into constructive contributions toward earthquake forecasting.

Inverse problems

Solving often nonlinear and ill-posed inverse problems forms another popular category of applications of deep learning in seismology. Deep learning for estimating physical parameters from observed seismic data can be divided into two main categories: (i) characterizing Earth's subsurface properties and (ii) characterizing the earthquake source.

Subsurface characterization

Various characteristics of DNNs have proven effective for solving inverse problems in seismology. Their ability to learn highly nonlinear relationships is the main idea behind fully data-driven end-to-end approaches for inferring a wide variety of subsurface properties such as seismic velocity (91–93), impedance (94, 95), petrophysical attributes (96, 97), dynamic reservoir properties (98, 99), groundwater distribution (100), and anelastic attenuation (101) directly from either pre-stack (92) or post-stack (102) seismic data.

Deep-learning applications for seismic inversion are not limited to end-to-end mappings. In conventional seismic inversion, deep learning can be used to learn an optimization algorithm (103), to introduce data-driven elements into an optimization procedure (104), or to replace computationally expensive steps (105). The representation-learning and dimensionality-reduction capabilities of DNNs are valuable attributes for seismic inversion. In full-waveform inversion (FWI), these characteristics are used to transform the large inverse problems by either skeletonizing the data (106) or reparametrizing the Earth model (107, 108) as the connection weights and/or learnable kernels in a DNN. Although reducing the dimension of observed and synthetic seismic data by use of DNNs can reduce the nonlinearity of the inverse problem and the risk of converging to a local minimum (106), the provided sparse representation of an Earth model in a lower-dimensional latent space reduces the computational cost and effectively regularizes the problem (103, 109). Representing Earth models with sparse multiscale feature maps through a DNN acts like a regularization technique by constraining the updates to the effective dominant features that represent the velocity model (107, 109). In addition, the differentiable representation of the Earth model facilitates the inclusion of physics-based constraints (95, 110). The flexibility of the open-source deep-learning platforms (such as Tensorflow or PyTorch) and their appealing computational capabilities (such as automatic differentiation and GPU parallel computing) motivate reformulating the entire FWI into a DNN with forward modeling guided by physics laws through neuron computations (111). A more advanced example of this is the development of specialized deep-learning architectures that are capable of learning the internal mechanism of general wave-based inversions (112).

These studies demonstrate the exceptional performance of deep-learning techniques in providing fast and reliable solutions to large-scale seismic inverse problems. Moreover, the processing time for both training and inference of deep-learning models can be an order of magnitude less than optimization time of classical seismic inversion approaches. Deep-learning inversion methods also tend to be less sensitive to noise (113) and missing low-frequency components (110) compared with thoses of classic FWI. They either are not sensitive to the choice of initial velocity models or do not require such initialization (91, 93). Last, the modularity of deep learning enables the processing of multimodal input data such as well-log and seismic or gravity and electromagnetic data, which facilitates joint inversion (114).

Earthquake source parameterization

Deep learning has great potential to provide near–real-time earthquake source parameters, such as hypocenters, magnitudes, and source mechanisms. Deep learning is effective for end-to-end learning of earthquake source properties based on waveform patterns propagating across a network of seismic stations (115–117). It can also be used to go beyond conventional multistation approaches by characterizing the source parameters from single-station data (118, 119).

Although multistation approaches are aligned with conventional methods, to date they suffer from smaller training datasets (usually a few tens of thousands of events). Single-station

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approaches are in an experimental phase that has explored the feasibility of maximizing the information that can be extracted from smaller units of observation (short windows of waveforms recorded by a single instrument) by using the feature learning and high nonlinearity of DNNs. The latter approach has the potential to enable monitoring of smaller events by using limited observations (or sparse network coverage) and benefits from much larger training datasets (hundreds of thousands of single-station observations) available for training deeper DNNs.

Overall, deep-learning applications for source characterization are in an earlier stage than those for subsurface inversion or automatic data processing. They encounter the challenges of the unavailability of genuine ground truth and the scarcity of large events (earthquakes of 7.0 magnitude or greater) in the training set. Incorporating the aleatory and epistemic uncertainties into the modeling procedure and physical constraints through physics-guided (*120*) or physics-informed (*121*) DNNs could be a promising direction to overcome these limitations.

Exploratory data analyses

Most applications of deep learning in seismology have been related to automation and modeling tasks; however, deep learning can be used in discovery mode to find patterns and insight in seismic data as well. Seydoux *et al.* (*2*) developed an unsupervised framework that combines a deep scattering network and a Gaussian mixture model to identify patterns in continuous seismic recordings through segment clustering. They used this to reveal the repeating precursory seismicity preceding a landslide in Greenland. This method has also been used for clustering of daily quasi-periodicity and numerous transient microevents on continuous seismic data recorded on Mars (*122*). A similar approach has been used to cluster urban seismic noise in Long Beach to identify their sources (*123*). In this approach, features that represent distinct clusters most effectively are extracted from input data by using a deep CNN through simultaneous optimization of clustering and feature-learning tasks. Unsupervised deep learning holds great potential for uncovering new insights in continuous seismic data. The intelligent feature extraction and dimensional reduction mechanism of this new group of DNNs (*19*, *124*) render them well suited for exploratory analyses of high-dimensional seismic data such as DAS.

Beyond conventional seismology

Seismology has many unconventional applications. The sensitivity of seismic sensing to the Doppler effect makes it an effective approach for the monitoring of moving objects such as humans, vehicles, and aircraft (*125*). Deep learning has also been used to detect human footsteps (*126*, *127*), discriminate between different classes of targets (such as humans versus vehicles) (*128*, *129*) and different types of vehicles (such as light vehicles, heavy vehicles, wheeled vehicles, and tracked vehicles) (*130*), or discriminate between different wildlife species and their behavior (*131*).

These unconventional applications of seismic sensing bring challenges and motivate technical approaches that could be of interest to the larger seismological community. For example, Bin *et al.* (*128*) showed that through on-site feature learning and classification by using CNNs to compress data and only transmitting the classification result to the server, it is possible to obtain similar results to those of state-of-the-art cloud-based deep-learning methods at only 1/10 of the computational time. Similar approaches could help reduce current challenges in seismic monitoring by use of fiber optic cables. Compressed sensing of seismic data (*132*), fusion of multisensor data (*129*), and federated learning (*133*) are other areas in which deep-learning techniques could assist in processing large data volumes.

The Nobel prize–winning detections of gravitational waves and subsequent groundbreaking discoveries made by using the advanced Laser Interferometer Gravitational-wave Observatory (LIGO) were made possible through a series of multidisciplinary research efforts. The advances in deep-learning seismology discussed above have the potential to contribute to gravity-wave measurements in multiple ways. Despite sophisticated seismic isolation systems, ground-based gravitational wave interferometers remain susceptible to ground shaking from large teleseismic earthquakes, which are one of the primary sources of interruption of the proper functioning of the interferometers and substantially reduce their duty cycle (*134*). Machine learning for ground motion characterization and EEW could help reduce that impact. Deep learning can also be used to predict on the basis of archival seismic data the ground-motion level and state of the gravitational wave detector during the event of an earthquake (*135*).

The template-matching algorithms used for the detection of gravitational waves can be complex and computationally expensive. Similar to earthquake signal detection and discrimination, deep learning has been shown to be an effective alternative for efficient detection and classification of gravitational-wave signals from different cosmic sources (*136*). DNN models can detect and estimate the true parameters of real events observed by LIGO with comparable sensitivity and lower error than that of template-matching, while being several orders of magnitude faster and more resilient to transient noise artifacts (*136*). These gravitational waves are extremely weak signals embedded in highly non-Gaussian and nonstationary noise owing to instrumental artifacts and environmental contamination. Denoising by use of deep-learning approaches and with the supervision of accompanied recorded seismograms is another approach for the detection of new events below the noise floor (*137*).

Discussion

Meta-analysis

To date, automation of seismic data processing and solving of inverse problems form the majority of deep-learning applications in seismology (Fig. 3). Although deep learning can be used to recognize patterns and extract insights from high-dimensional seismic data, their application for exploratory analysis of seismic data and scientific findings are largely unexplored (Fig. 4A). This might be due to an atmosphere of skepticism regarding the black box nature of deep-learning models. In most seismic data-processing tasks, however, the cost of a wrong answer is low relative to the value of a correct answer. For example, the risk of picking wrong or inaccurate phase arrivals in earthquake monitoring is low because wrong or inaccurate arrivals will usually be eliminated or refined in subsequent processing. Thus, having a deep-learning picker that can do this task with high efficiency but some brittleness, which is defined as the tendency to get fooled (*138*), and imperfect precision does not undermine the utility of the method. Even in applications such as EEW, for which the cost of a wrong answer may be high, deep learning may offer the best available approach (*76*). Last, although deep learning cannot be used to establish causation or understand processes directly from data, it can be used to inspire and direct researchers toward scientific discoveries.

The scarcity of large-scale labeled field data for model building is more evident in active seismology and for applications such as seismic image processing, interpretation, and subsurface characterization (Fig. 3). This scarcity is due to both the limited number of large-scale and publicly available datasets and the subjectivity of labeling for these tasks. Earthquake signal processing tasks—such as seismic phase picking, phase association, and earthquake detection—benefit from larger and publicly available training datasets.

Supervised learning has dominated deep-learning approaches used in seismology to date. However, important applications for alternative unsupervised and semisupervised approaches are being increasingly found. Unsupervised learning has focused on seismic facies analyses, seismic denoising, event discrimination, and exploratory data analyses. As the number of developed models for each seismological task grows, techniques such as ensemble learning are likely to find more applications.

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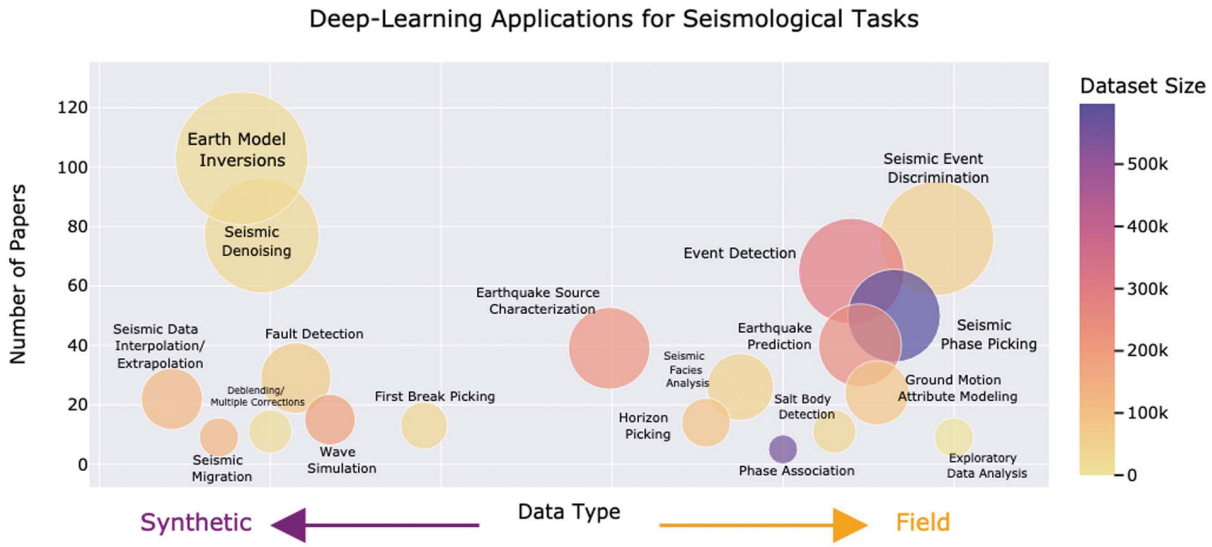

Fig. 3. Deep-learning applications for seismological tasks. The size of each circle is scaled to the number of published papers for each application and color-coded according to the average size of training data. Horizontal axes represent the average percentage of observational and recorded data versus synthetic training data.

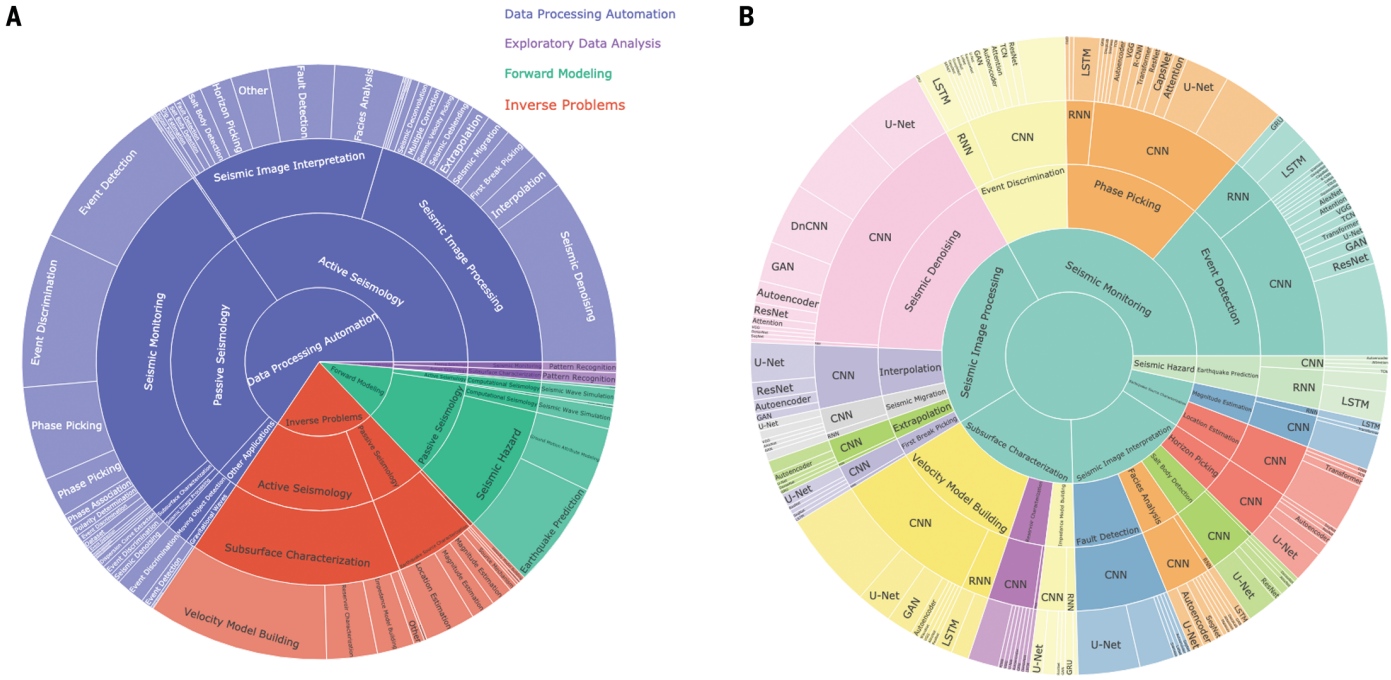

Fig. 4. Hierarchical distribution of studies in our database. (**A** and **B**) Studies are distributed according to (A) the seismological applications and (B) used DNNs.

Fine tuning of trained models, especially those trained with synthetic data, is popular in seismology, and we expect that meta-learning will be a growth area in seismology.

CNNs dominate seismological models across many sectors (Fig. 4B) because of their effectiveness for automatic feature extraction and sparse representation learning of seismic data. Some characteristics of seismic data such as their limited bandwidth, sequential nature, or data structure (for example, 3D and 4D imaging in active seismology) have motivated developments of specialized and hybrid DNN architectures composed of a mixture of neural-layer types. RNNs are popular in tasks in which temporal relations in data play a crucial role in the modeling. Network architectures with contracting encoders and expanding decoders such as U-Net and Autoencoders are highly suitable for modeling seismic data across many applications. GANs are another popular approach for seismic data processing. Among RNNs, long short-term memory (LSTM) units are the most popular network type. The sequential nature of seismic data makes RNNs appealing for many seismological tasks. More recently, neural networks with attention mechanisms, dilated CNNs, and physics-informed DNNs are gaining momentum.

Explainable deep learning

Despite the widespread application of deep learning in seismology and its breakthrough performance in multiple areas, the black-box nature remains a source of skepticism. Accepting indecipherable models on their own terms

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can contribute to scientific goals of the community (139), but presenting the properties of these models in understandable terms to humans (interpretability) and revealing the underlying reasons for their output decision (explainability) (140) would allay such skepticism and lead to wider adoption and new insights.

Currently, most deep-learning approaches in seismology concern building a model based on known input-output pairs to predict the outputs corresponding to previously unseen inputs without interpretability or explainability. In most of these cases, it is not possible to explain why specific outcomes were obtained from an algorithmic point of view nor what else these models can tell us.

Visualizing the learned kernels at different layers of a CNN is the main approach used in some of the seismological studies (93, 107, 141, 142) to understand something about the representations a deep learning model has learned and how it drives the output; however, the extracted multiscale feature representations from seismic data are too complex for direct visualization to provide interpretable and understandable insights.

Although it may not be possible to elucidate how a deep-learning model works, gaining useful information about learned models is possible by using interpretation tools such as backward propagation techniques, saliency maps, or heatmaps to reveal relevant patterns in the input based on feature importance or relevance scores (143, 144). These interpretable representations of the input can be used along with domain knowledge to explain the output.

Mousavi *et al.* (12), for example, used attention mechanisms in their detection and picker DNN. They used a hierarchy of attention modules to gain insight into the interaction of task-specific decoder branches and where and on what the DNN focused. Doing so gave a degree of interpretability to their results. Attention was used in (54) to build an interpretable deep-learning model for seismic facies analysis. They used 3D spatial-spectral attention maps to reveal the relations between the seismic spectral response (input) and the geological formations (output). Liao *et al.* (22) used a backward propagation technique to compare different deep-learning models for phase-picking on the basis of the learned representations from data.

In addition to explaining the outcomes, designing an interpretable deep-learning model is possible in which the model itself and its components became explainable by using domain knowledge (111, 145). In this approach, the network architecture is designed to provide insight by promoting or restricting specific attributes of the data being modeled (146, 147). Zhu *et al.* (87), for example, used a mixture of Gaussian diffusion kernels to build an interpretable model of anisotropic spatial dependency in seismicity patterns. The parameters of these kernels are reparameterized by DNNs and learned from training data by using an imitation learning approach. Such an interpretable model can help uncover patterns [and hopefully causal relationships (148)] in data.

Incorporating geophysics into DNNs

Deep-learning approaches that incorporate physical laws have gained momentum in the machine learning community (149) and a growing number of implementations in seismology. The main idea is to integrate data and mathematical physics (domain knowledge) models, even if only partially understood. The key objectives are to provide interpretability and explainability for a deep-learning method, to regularize them in the presence of missing or noisy data, and to enhance their out-of-distribution generalization by restricting the solution space to physically plausible solutions.

Domain knowledge can be incorporated into the training data, the hypothesis space, and the training procedure of a deep-learning model (Fig. 5). Among these, incorporating the domain knowledge into training data through data generation and augmentation by using numerical simulations is the most commonly used approach in seismology for active seismic data processing (47, 150), wave simulation (68), and velocity estimation (91). These models are usually fine tuned through some examples of field data.

The hypothesis space can be constrained with domain knowledge through design of specialized neural network architectures based on the characteristics of the problem. Such physics-inspired neural network architectures have been developed for seismic FWI (111, 145). Sun *et al.* (111) used a recurrent architecture to model time-dependent dynamics of seismic wave propagation. To enforce physical rules for wave propagation in the model, they set up LSTM-like cells—including some physical operators, which take the wavefield at a past instant as input—estimated the shot record at the current instant as output, and saved the modeled wavefield in the memory cell for calculating the next time step. In these models, the trainable parameters of the DNN can be viewed as a reparameterization of an Earth model. The forward propagation of information through this representation resembles the forward propagation of a seismic wavefield through a heterogeneous medium. Thus, training such a network and updating its weights by using DNN back-propagation and based on the misfit between the simulated and observed seismic data amounts to back-propagation of the residual wavefield and under certain assumptions is equivalent to the optimization process in conventional gradient-based FWI. This can result in a more interpretable model in which the latent representation is physically meaningful.

Physical constraints can be introduced into the training process either by incorporating them into the loss function (implicit approach), by explicit inclusion of a numerical modeling into the training loop, or both. The first approach is used more recently in seismology and primarily for forward problems such as simulation of the pressure wavefield (68) or estimation of first arrival times (77). In these studies, a form of the wave equation (68) or Eikonal equation (77) is used to penalize the

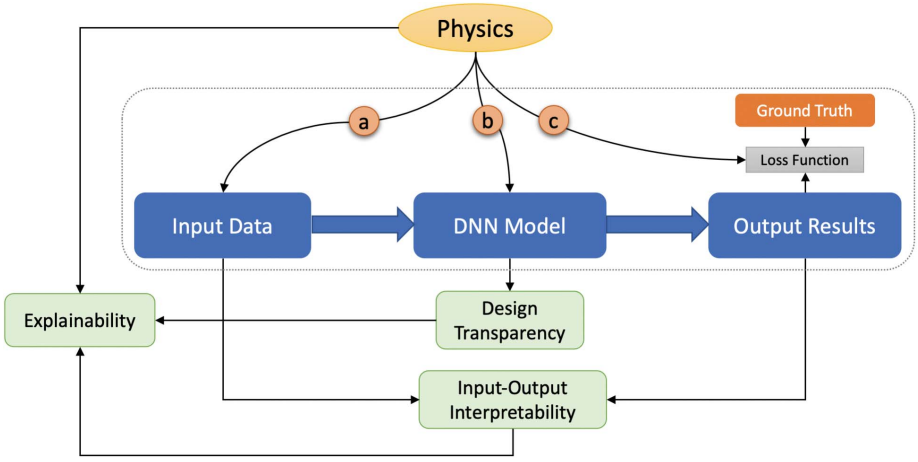

Fig. 5. Explainable DNN. The domain knowledge, physics, can be incorporated into the training data through (**A**) data simulation, (**B**) the hypothesis space by designing specialized neural network architectures, and (**C**) the training procedure of a deep-learning model by enforcing consistency with governing equations. The architecture design based on domain knowledge provides some transparency and intepretability of the model. The interpretable representation of the input data along with some domain knowledge can be used to explain the output.

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loss functions used for training the DNNs. Thus, during the training the model is simultaneously optimized to fit the observed data while respecting physical constraints. The examples of the second approach are mostly in seismic FWI in which a physics-based forward modeling of synthetic seismograms is performed as a part of the training loop to embed some physics-based constraints in the model (*95*, *107*, *108*, *110*). In this approach, the mismatch between observed and numerically simulated seismic data is used to compute the gradient of the misfit function with respect to the DNN weights and update the initial Earth model represented by a DNN.

Physics-guided (or physics-informed) deep learning is effective for solving ill-posed inverse problems as well as for forward problems in seismology. They hold a great potential for wider applications in seismology beyond examples mentioned above. Developing hybrid approaches by combining multiple physical constraints (for example, simultaneously in architecture design and loss function), using physics-based GANs to enforce realism for data generation (*151*), constraining a model statistically (*151*), quantifying uncertainties (*152*), and learning physical operators (*153*) are other directions of interest.

Uncertainty quantification

As development of deep-learning models for some seismological tasks progresses toward routine operations, quantification of corresponding aleatory and epistemic uncertainties becomes essential to assess the reliability of output predictions. Bayesian DNNs can play a key role in this area by providing probabilistic treatment of neural networks (*154*). In these models, a prior probability distribution is placed over DNN parameters, and a posterior of these parameters is learned during the training. This strategy can be seen as learning by an infinite number of neural networks given the data in which the inference occurs by aggregating the results through ensembling. The high time complexity of Bayesian DNNs is the main challenge in their practical application; however, advances in scalable and easy-to-implement adaption of Bayesian DNNs are set to change that (*155*–*158*).

Monte Carlo dropout sampling (*158*) is a popular technique for uncertainty estimation because of its flexibility and ease of implementation (*57*, *94*, *118*, *159*). Mousavi and Beroza (*118*), for example, developed a Bayesian CNN for earthquake location by using a customized loss function to estimate the aleatory uncertainties directly from data during the training process. The epistemic (model) uncertainties were estimated through dropout during inference. In this approach, the loss function acts like an intelligent loss function and reduces the effect of uncertain data, whether because of noise or unreliable labels; however, the effectiveness of this approach in estimating model uncertainties is limited.

Recent computational advances (*160*, *161*) for building deep probabilistic models provide alternative options for a more robust estimation of model uncertainties. Feng *et al.* (*56*) developed a Bayesian deep-learning framework for seismic facies analysis in which the Bayesian variational inference for approximating the posterior distribution of DNN parameters is formulated as an optimization process. In this approach, multiple random samples are drawn from the estimated posterior distribution to quantify the model uncertainties by using the prediction entropy.

Besides uncertainty quantification, the Bayesian DNN approach has some implicit regularization effects that can suppress overfitting, especially when there is insufficient data. We expect that there will be an increased effort to develop probabilistic deep-learning models, especially for tasks such as seismic image interpretation, in which the cost of a wrong decision is high. Developing physics-informed Bayesian models would be one approach in this direction.

Training data limitations

The existence of large-scale realistic training datasets with diverse characteristics and precise labels is crucial for the success of deep learning. Uncertainties and nonuniqueness of manual labels owing to limited resolution, presence of noise, different levels of expertise, cognitive biases, and inherent ambiguity of tasks is a limiting factor in the development of training data in seismology. Two analysts may disagree about the existence or location of a particular fault, or a deep horizon in a seismic image, or arrival times in an earthquake signal when labeling them, which biases training data. A challenging task in building training datasets is quality control of the labels.

In addition to labeling uncertainties, some tasks such as seismic velocity inversion, image processing, and interpretation face the challenge of a scarcity of public large-scale labeled data from field observations. One approach to address this limitation is to generate physically realistic synthetic data that represent a variety of complex geological settings (*162*). For seismic velocity inversion, Wang *et al.* (*163*) used smoothed versions of natural images as seismic velocity models to produce synthetic seismic traces to use as training data. Feng *et al.* (*164*) developed a real-time style transform method to create a large set of physically realistic subsurface velocity maps from natural images. They used these data in their multiscale deep learning–based FWI. Araya-Polo *et al.* (*165*) used GANs to generate a large number of geologically feasible models to augment their original training data set.

Another approach uses inductive transfer learning. This was used to fine-tune pretrained CNN models on handwritten images (*8*) or natural images (*166*) to classify volcano-seismic events according to their time-frequency representations. Developing semisupervised (*20*, *95*, *167*) and unsupervised (*19*, *35*, *110*) approaches for reducing a need for labels is a growing trend. Other options are adding constraints to the training procedure (*167*, *168*) and reducing the degrees of freedom (the number of trainable parameters) in supervised approaches by incorporating physics-based constraints.

Out-of-distribution generalization

To evaluate the quality of the obtained model in most seismological applications, authors usually perform a blind test on a subset of the data not used in training and/or samples of data from other datasets (or regions). A blind-test dataset that is representative of the full diversity of the real world is needed to evaluate the generalization of a model robustly. Statistical significance of such tests is usually not presented, and it is unclear to what extent a trained network can generalize.

For applications such as earthquake signal detection and phase picking, in which the model generalizations have been studied extensively (*12*, *22*, *23*, *25*, *29*, *169*), neural network architecture, neural network type, and the size of a training set have been shown to be not the only defining factors for good out-of-distribution generalization (*12*) and that hyperparameter tuning (*29*), training procedure, and data augmentation (*170*) also play important roles. These effects have seen less attention in seismic image denoising, processing, and interpretation, in which model building is often done by using the synthetic data and data augmentation is limited (in part because some common image augmentation techniques, such as rotation, result in physically implausible models).

Generalization of a deep-learning model always has a limit because each model may have blind spots, and out-of-distribution cases can always arise. This makes it difficult to have a distinct and robust measure of the generalizability. It remains unclear how much improvement might be obtained by using physics-guided DNNs or Bayesian frameworks. More systematic studies are required to understand and characterize this important property of deep-learning models.

Benchmarking pitfalls and concerning trends

Despite the recent increase of applications of deep learning in seismology, not all the reported improvements lead to real-world performance increases. Often it is not clear what worked and why or which deep-learning approaches actually lead to better performance.

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The choice of weak or often not properly fine-tuned baselines, different datasets, data preprocessing, evaluation protocols, performance metrics, the lack of result reproducibility, ablation studies, error analysis, and robustness checks are all factors that make it difficult to assess progress. To overcome this concerning trend and avoid the "phantom progress" phenomena observed in other fields (*171*), encouragement and ultimately enforcement of standard benchmarking need to be a necessary foundation of empirical machine learning research.

The implementation of benchmarking is currently diverse and inconsistent across deep-learning seismology. Applications such as subsurface characterization have well-established benchmarks, such as the Marmousi (*172*) or BP2004 (*173*) models that are widely used; but even in this case, performance evaluation can be qualitative. Applications such as event discrimination and seismic denoising have seen a relatively better adaptation of reporting metrics (for example, precision and recall, SNR, or mean-squared error) but suffer from the lack of canonical benchmarks. Because this is still a young field, there is no standard benchmark or evaluation setup for most of the deep-learning applications in seismology. As a result, researchers choose their own test data and evaluation criteria to showcase their method and (perhaps unknowingly) tend to find good performance for their models. This lack of community-driven sets of benchmarks and evaluation guidelines hinders progress in the field.

Given the indispensable role of benchmarking for measuring progress, the community involved in deep-learning applications needs to set up more unified, standard, and rigorous evaluations that will ultimately lead to the development of insight and understanding because the community also are those who will endorse and enforce these practices. This requires first determining the guidelines for the reproducibility of the results, the choice of the baselines, building of benchmark datasets, and defining of task-specific metrics and standards for reporting the results. When creating a benchmark dataset for machine learning applications, documentation of its composition, collection process, labeling uncertainties, and applicable tasks, among other criteria, are all important. Only a few such benchmarks currently exist in seismology (*174*–*177*) specifically designed for machine learning. Evaluating a model on multiple benchmarks clearly would provide a more robust performance measurement. In this spirit, setting up guidelines on how to use benchmarks is also important. Clear reporting of the training data and its size, model size, time and memory complexities, inference latency, suitable metrics indicating model variance, its generalization to the out-of-distribution data, amount of hyperparameter tuning, and a discussion of statistical significance of the results are all important factors that have received too little attention in the seismological community. Setting up clear guidelines for reviewers is the next step that could help promote and enforce greater rigor in the field and render more accurate and fair judgments when comparing models.

Conclusions

Deep learning has entered almost every subfield of seismology and has the potential to dominate algorithm development. Yet as a general-purpose tool for scientific discovery, deep learning is in its infancy. Investing more effort on physics-based and explainable deep learning will help the field to mature. Research is rapidly moving from proof-of-concept efforts to real-world applications, and estimation of uncertainty will become essential. Given common pitfalls and concerning trends, setting up standard benchmarking and evaluation protocols is critical. An understanding of the limitations and breaking points of these models will be extremely helpful. Despite these challenges, deep-learning approaches will clearly contribute substantively to seismology by optimizing results, providing value, and sparking inspiration. Given the incredible ability of DNNs for modeling and processing large-volume, high-dimensional seismic data and their remarkable performance in varied seismological tasks, deep learning will soon become the norm in multiple seismic data analyses, notwithstanding issues of interpretability and brittleness. Deep learning will not replace expert seismological analysts, but those analysts who use it will outperform those who do not for many types of problems.

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Deep-learning seismology

S. Mostafa Mousavi and Gregory C. Beroza

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Large-scale learning

The large amount and availability of datasets in seismology creates a great opportunity to apply machine learning and artificial intelligence to data processing. Mousavi and Beroza provide a comprehensive review of the deep-learning techniques being applied to seismic datasets, covering approaches, limitations, and opportunities. The trends in data processing and analysis can be instructive for geoscience and other research areas more broadly. —BG

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